Botulinum A Toxin (BOTOX) in the Lower Eyelid: Dose-Finding Study

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BACKGROUND. **Botulinum toxin type A (BTX-A, BOTOX) is an excellent therapeutic option for hyperkinetic facial lines. It improves wrinkles by relaxing the muscles of facial expression, which underlie the rhytids. Periocular wrinkles such as lateral orbital rhytids respond well to treatment. BOTOX can be used in the lower eyelid to improve wrinkles and widen the eye.**

OBJECTIVE. **To determine whether there is additional benefit in using more than 2 U of BTX-A to improve infraorbital wrinkles and widen the eye.**

METHODS. **Nineteen women had BTX-A injected into the orbicularis oculi muscle. Eleven women had 4 U injected into the lower eyelid bilaterally, 3 mm below the ciliary margin, and 12 U of BTX-A injected into one lateral orbital (crow's feet) area. Eight patients had 8 U injected bilaterally into the lower eyelid and 12 U placed unilaterally into the crow's feet. Physicians and patients independently evaluated the degree of improvement (grade 3 = dramatic improvement, grade 2 = moderate improvement, grade 1 = mild improvement, and grade 0 = no improvement). Single investigator analysis was used to measure, in actual millimeters, the amount of increase in palpebral aperture. Side effects were noted.**

RESULTS. **Improvement was noted in lower eyelid wrinkles by both physicians and patients at both dose groups. When only the lower lid was injected, patients reported an improvement of 1.18 with 4 U and a grade of 1.63 with 8 U. When both the lower eyelid the lateral orbital area were treated, an improvement of 1.73 was reported with 4 U and a grade of 2.25 reported with 8 U in the lower eyelid. Physician evaluations had grades of 1.85 for 4 U alone and 1.85 with 8 U alone. Grades of 2.35 and 2.25 were obtained for 4 U plus 12 U and 8 U plus 12 U, respectively. An increase in palpebral aperture (IPA) occurred in all subjects. Subjects who received 4 U in the lower eyelid alone had a 1.8-mm IPA at rest and a 2.6-mm increase at full smile. Subjects who received 8 U of BTX-A alone in the lower lid had an IPA of 2.2 mm at rest and 2.9 mm at full smile. Eyes treated with 12 U in the bilateral orbital area plus 4 U in the lower eyelid had an IPA of 2.2 at rest and 4.5 mm at full smile. Those treated with 8 U in the lower lid plus the crow's feet had an IPA of 1.5 at rest and 4.0 at full smile. Side effects increased with dosage, with eight of eight subjects in the 8-U dose groups reporting bothersome side effects such as lower eyelid edema and incomplete sphincter function.**

CONCLUSION. **A dose–response curve is seen with increasing doses of BTX-A used in the lower eyelid. Treatment of the lateral orbital area in combination with the lower lid produces a synergistic response at lower doses, but at higher doses, a plateau effect is suggested. Although increasing doses of BTX-A increases eye widening, unattractive results and side effects are seen at higher doses. The authors recommend that lower 2- or 4-U doses of BTX-A be used in the lower eyelid and specifically discuss techniques.**

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BOTULINUM-A EXOTOXIN (BTX-A, BOTOX) is used extensively in cosmetic dermatology.[1–3] It improves wrinkles by affecting the underlying muscles, which contribute to the overlying rhytides.[4,5] The toxin's effect is temporary and reversible. Botulinum toxin has been widely used in the upper third of the face,[6] especially the glabella.[7–9] BOTOX Cosmetic holds Food and Drug Administration approval for the treatment of glabellar frown lines.[10] Both BOTOX and BOTOX Cosmetic have been used to treat other facial wrinkles such as the forehead. Wrinkles that overlie the nasalis muscle, pruney chin, marionette lines, and platysmal bands[11,12] can be helped. Eyebrows can be raised with BTX-A.[13] Periorbital rhytides respond well to BTX-A.[14,15]

We have previously reported on the use of BTX-A in the lower eyelid.[1] Our studies have shown that 2 U of

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botulinum toxin in the lower lid improves infraorbital wrinkles and widens the eye. The effect is more noticeable when the lower eyelid is treated in conjunction with the lateral orbital (crow's foot) area. Many patients with narrow palpebral apertures find the eye widening attractive as it gives the lower eyelid a more rounded appearance. We report herein a dose-finding study using additional units of BTX-A in the lower eyelid.

Methods

Our methods have been extensively detailed in our previous publication.[1] Briefly, women 18 to 60 years older having a hypertrophic orbicularis oculi muscle of the lower lid contributing to lower eyelid folds or rhytides were candidates for the study. Patients were evaluated to make sure that they had no dry-eye symptoms and no existing ectropion. They had to have an acceptable snap test and no evidence of scleral show. All patients could not have had botulinum toxin treatment of the lateral orbital area for 6 months or longer and had to have been BTX-A naive in the infraorbital area. Exclusion criteria were as follows: previous lower eyelid surgery, periocular resurfacing, ectropion, eyelid asymmetry that would interfere with assessment, allergy to BTX-A, neurologic or neuromuscular disease, pregnancy, or lactation.

Patients were evaluated regarding their suitability for the study. A urine pregnancy test was done to ensure that they were not pregnant. A snap test was performed. Assessment of the lower eyelid folds and rhytides was done, and the patient was photographed preoperatively with the head in a neutral position. Patients were photographed using Ektachrome 100S (Kodak, Inc., Rochester, NY) transparency film at a 1:5 magnification. A millimeter scale was taped to the patient's facial skin as a control. Patients were photographed anteriorly and bilaterally at both rest and maximal smile. The mouth was included in the photographs to ensure a complete smile.

A modified Schirmer's test was performed by instilling 0.5% tetracaine ophthalmologic drops bilaterally. Schirmer's strips were placed in the lateral one third of the lower lid and tear production (in millimeters) measured after 60 seconds. Patients were randomized into 4- or 8-U lower lid dose groups.

BTX-A was reconstituted using a milliliter of preservative-free physiologic saline on the day of injection. This produced a concentration of 1 U per 0.01 cc. BD U-100 0.3 cc insulin syringes equipped with an Ultrafine II 30-gauge needle (Product #328438; Benton Dickinson Corporation, Rutherford, NJ) were used to inject BTX-A.[16]

Two injection points in the lower eyelid were used for the 4- and 8-U dose groups, as shown in Figure 1. With the syringe horizontal to the ground, a lateral approach was made, and the needle was inserted just lateral to the midpupillary point 3 mm inferior to the lash margin. The tip of the needle was advanced to the midpupillary point, and either 2 or 4 U of BTX-A (0.02 and 0.04 mL) were placed just subdermally. A second injection point halfway beneath the midpupillary line and the lateral canthus 3 mm inferior to the ciliary margin was used to inject the other 2 or 4 U. The injections were chosen near the ciliary margin to effect orbicularis oculi alone. For lateral crow's feet injections, patients were randomized using the day of their birth. Patients having an even day of birth were injected on the right lateral orbital area, and those with an odd day of birth were injected on the left. Three injection points were chosen, each at a 1.5-cm radius from the lateral canthus, which is 1 cm

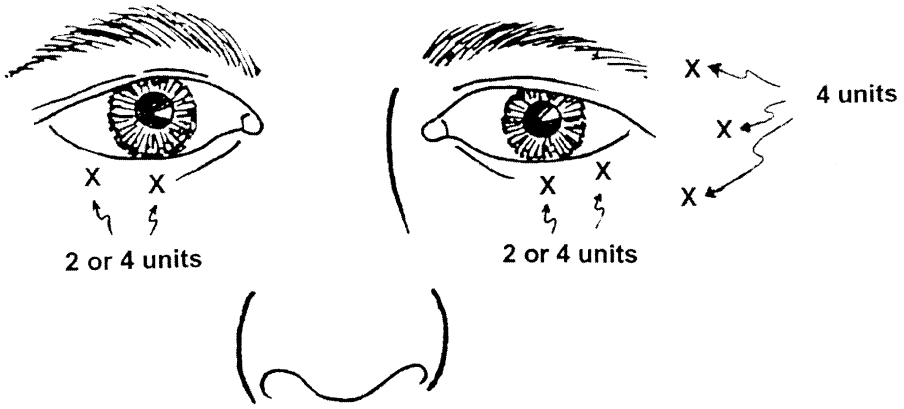


Figure 1. Study design indicating placement of BTX-A injections. Either 4 or 8 U of BTX-A were injected just subdermally in the lower eyelid in two equally divided doses 3 mm below the ciliary margin. The lateral orbital (crow's feet) area was treated with three subdermal injections of 4 U each 1 cm outside of the orbital rim for a total dose of 12 U.

outside of the orbital rim. Each injection point received 4 U of BTX-A (0.04 mL) subdermally for a total of 12 U. Patients received written postoperative instructions.

Patients were seen 2 weeks after the baseline visit. Standardized photographs were repeated. Both patients and investigators were asked to evaluate the degree of improvement in each lower lid independently on a 0 to 3 scale (0 = no improvement, 1 = mild improvement, 2 = moderate improvement, and 3 = dramatic improvement). Physician evaluators were blinded to the patient's assessment. If the patient had complained of dry-eye symptoms, the modified Schirmer's test was repeated.

After these assessments and photographs were completed, the subjects exited the study. BTX-A treatment of the untreated lateral orbital area was offered to the participants to return them to a more symmetrical appearance.

Results

Data on 34 patients are reported in this article. Fifteen patients were able to complete our original 2-U infraorbital study. Eleven patients were able to complete the 4-U dose range, and eight patients were treated with 8 U in the lower eyelid.

Patients and physicians assessed the degree of improvement independently using the rating scale. Data from the 4-U group are reported in Table 1. The table reports the degree of improvement from 4 U of BTX-A being placed in the lower lid as well as 4 U in the lower lid plus 12 U of BOTOX in the lateral orbital "crow's feet" area. The degree of improvement placing 4 U of BTX-A in the lower lid had an average patient assessment grade of 1.18. On the side that received 4 U in the lower lid and additional 12 U of BOTOX in the lateral orbicularis area, the patients reported an average grade of 1.73. Thus, patients with 4 U in the lower lid reported slightly above mild improvement and those that received 4 U in the lower lid and 12 U in the lateral area reported a value approaching moderate improvement.

The physician's assessment of degree of improvement was done independent of the patient's assessment. The physician's assessment of an isolated 4 U of BTX-A used in the lower lid reported an improvement of 1.85, while the physician's assessment of treating both the lower lid and the lateral orbital area reported an improvement scale of 2.35.

Table 2 shows the patient's and physician's assessment of improvement when using 8 U of BTX-A placed subdermally in the lower lid along with 8 U in the lower lid plus 12 U in the lateral orbicularis. The patient evaluation of 8 U placed solely in the lower lid reported a degree of improvement of 1.63. The patients felt that 8 U placed in the lower lid plus 12 U in the lateral orbital area reported a degree of improvement of 2.25. Independent physician evaluation reported an improvement scale rating of 1.88 when 8 U was used alone in the lower lid and a degree of improvement of 2.25 when 8 U was placed into the lower lid along with 12 U in the lateral orbital area. Figure 2 diagrams the results of improvement, as rated by both patient evaluation and physician evaluation.

Measurement of the increase in palpebral aperture was undertaken when using the photographic transparencies. This was done via the use of a photographic

Table 1. Degree of Improvement in the 4-U Dose Group

	Patient Evaluation		*Physician Evaluation*	
	4 U Lower Eyelid	*4 U Lower Eyelid and 12 U Crow's Feet*	*4 U Lower Eyelid*	*4 U Lower and Eyelid and 12 U Crow's Feet*
E.B.	1	1	2	2
K.H.	3	3	3	3
J.L.	0	0	2	2
T.M.	0	1	1	2
A.M.	3	3	3	3
S.P.	3	3	3	3
L.P.	0	1	Not evaluated	Not evaluated
C.S.	1	3	1	2
A.W.	0	1	0.5	1.5
K.W.	1	2	1	2
I.Z.	113/11	119/11	218.5/10	323.5/10
Average	1.18	1.73	1.85	2.35

Rating scale: 0 = no improvement; 1 = moderate improvement; 2 = moderate improvement; and 3 = dramatic improvement.

Table 2. Degree of Improvement in the 8-U Dose Group

	Patient Evaluation		*Physician Evaluation*	
	8-U Lower Eyelid	*8-U Lower Eyelid 12-U Crow's Feet*	*8-U Lower Eyelid*	*8-U Lower Eyelid and 12-U Crow's Feet*
J.E.	2	2	2	2
K.W.	1.5	2	2	2
J.R.	2	3	3	2
N.K.	2	3	2	3
L.H.	2	3	2	3
T.B.	1.5	2	1	1
D.L.	2	3	2	3
M.J.	013/8	018/8	115/8	218/8
Average	1.63	2.25	1.88	2.25

Rating scale: 0 = no improvement; 1 = moderate improvement; 2 = moderate improvement; and 3 = dramatic improvement.

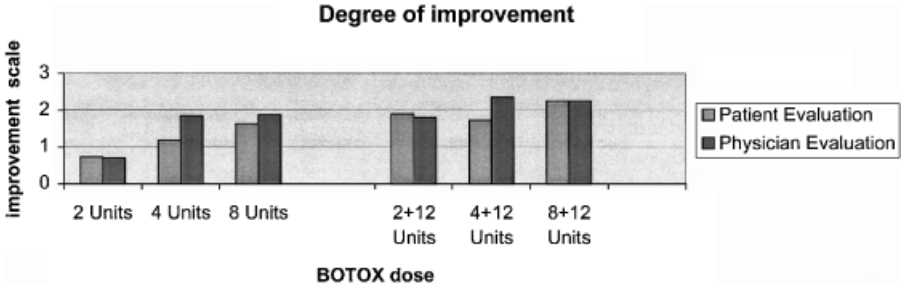


Figure 2. Ranking of improvement is shown on the *y* axis with a grade of 3 equals dramatic improvement, 2 equals moderate improvement, 1 equals mild improvement, and 0 equals no improvement. Both physician's and patient's assessments are shown.

enlarger equipped with an 80-mm lens used to project the transparencies. The enlarger was positioned so that the actual image was life size (the millimeter scale taped to the patient's cheek serving as an internal reference). Not all patients could be accurately evaluated because of technical errors such as incomplete eye opening on photography, incomplete smile, or inadvertent omission of the millimeter scale on the patient's cheek. The increase in palpebral aperture was measured in the midpupillary line and at rest and at "full smile." The results of the increase in palpebral aperture are reported in actual millimeters.

Figure 3 shows that those patients who received 4 U in the lower lid alone had a 1.8-mm increase in their resting palpebral aperture and a 2.6-mm increase in their palpebral aperture at full smile. The side receiving 4 U of BOTOX in the lower lid as well as 12 U in the lateral orbital area had a 2.2-mm increase in palpebral aperture at rest and an increase in palpebral aperture of 2.9 mm at full smile. Figure 4 shows good improvement. Those people receiving 8 U in the lower lid alone had an increase in palpebral aperture of 2.0 mm at rest and a 4.5-mm increase at full smile. Those patients who had received 8 U in the lower lid plus 12 U laterally had an increase in palpebral aperture of 1.5 and 4.0 mm at rest and full smile, respectively.

Adverse Events

Three of 11 patients in the 4-U dosing group reported the feeling of some dryness of the eyes. This was transient or mild in most patients. There was no ectropion, bruising, headache, diplopia, photophobia, or ptosis reported in this group. The lack of bruising may be partially explained by a slow injection technique and the use of the small insulin syringe integrally equipped with the sharp 30-gauge needle.[16] In the 8-U dose study group, more bothersome problems were noted in five of eight patients. Three of eight patients reported a transient lower eyelid edema with an increasing sense of lower eyelid "fullness." Photophobia in bright sunlight was noted by two of eight patients. Incomplete eyelid sphincter function was noted by two of eight patients. Specific comments included these: "my bags have dropped," "sun bothers me when I go outside," and "I cannot keep the soap out of my eyes when I wash my face" at the highest dosage group. Complete spontaneous comments regarding side

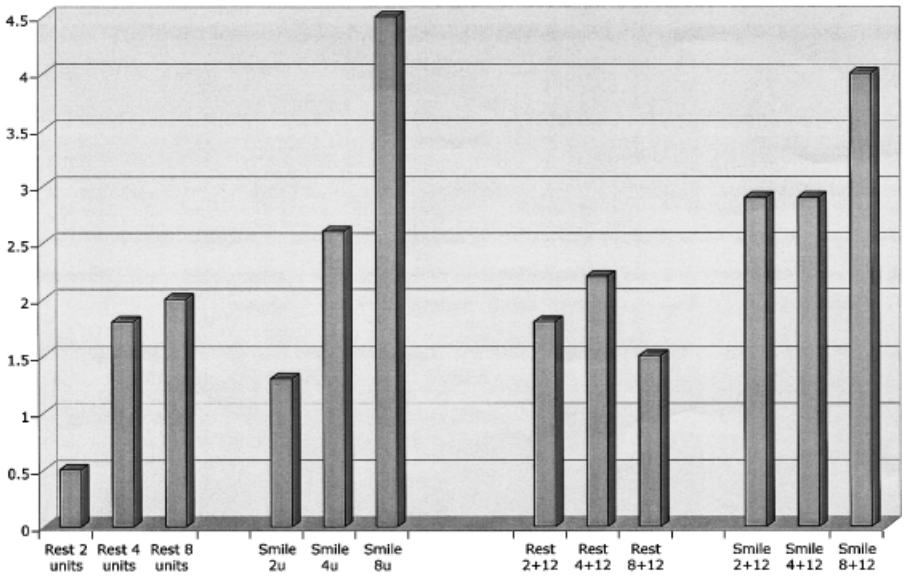


Figure 3. An excellent dose–response showing an increase in palpebral aperture (in mm) is shown both at rest and at full smile when 2, 4, or 8 U of BTX-A are used in the lower eyelid. The figure shows a synergistic response when 2 U is used in combination with 12 U in the lateral orbital (crow's feet) area. A synergistic response to 4 or 8 U plus treatment of the lateral orbital area is not as clearly shown.

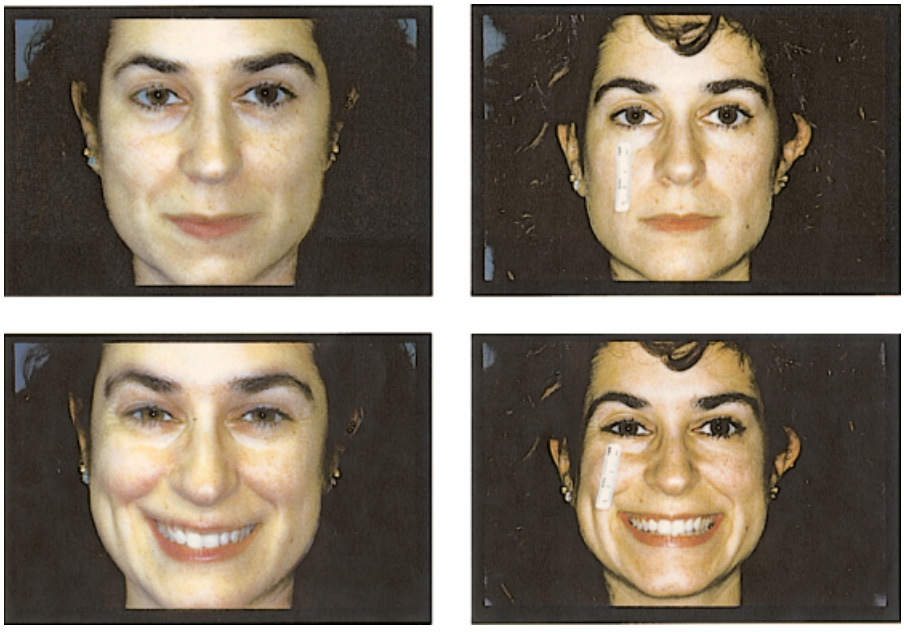


Figure 4. Good improvement of lower eyelid wrinkles plus eye widening is shown with 4 U in the lower eyelid. The patient's left side has had 12 U injected into the lateral orbital area.

effects of the treatment are reported in Table 3. Complete resolution of the adverse effects occurred within 3 months.

Discussion

This study has demonstrated a dose–response when using BTX-A subdermally 3 mm inferior to the lid margin in the lower eyelid. Data are presented in this article combining our data on 4- or 8-U dose groups with our previous 2-U dose group in the lower lid study.

Clinically, most patients who seek BOTOX therapy for improvement of wrinkles have multiple areas treated.[12] We previously demonstrated that one could safely add 2 U of BTX-A in the lower eyelid along with treatment of the crow's feet and achieve

Table 3. Patient Commentary About Side Effects of Treatment

2-U study group	No side effects noted.
4-U study group	Mild dry eyes noted in 3 of 11 patients. "Transient" or "mild" in most patients. One lasted only 1 week. One patient had to decrease contact lens use because of dry eye symptoms.
8-U study group	Problems noted in five of eight patients. Transient lower eyelid edema was noted by three of eight patients, photophobia in bright sunlight noted by two of eight patients, incomplete sphincter function of the eye noted by two of eight patients. Specific comments included these: "My bags have dropped." "Sun bothers me when I go outside." "I can't squint in the bright sunlight." "My lower lids are puffy and worse at the end of the day." "My lower lid stays down." "My eyes sting when water gets on my face." "I can't keep the soap out of my eyes."



Figure 5. Slight lower lid edema is shown when 4 U of BTX-A are used in the lower side. The left side with 12 U in the lateral orbicularis oculi appears to be slightly worse.

a moderate improvement in lower eyelid wrinkles without complications. In this dose–finding study, Figure 2 demonstrates a nice dose–response curve in the degree of improvement when 2, 4, or 8 U are used in the lower eyelid alone. Increasing improvement is noted with an increasing dose of botulinum toxin in the lower eyelid. This dose–response is also seen when 2, 4, or 8 U are used in combination with 12 U in the lateral crow's feet. We can improve slightly the degree of improvement when 4 or 8 U are used above 2 U, but a plateau effect is suggested. The duration of the complete response was not addressed in this study.

Subjects in the 2-U study group reported no negative side effects. An increase in side effects was noted when the amount of botulinum toxin was increased to 4 or 8 U. Five of eight patients in the 8-U study group were not happy with their treatment, reporting bothersome side effects detailed in Table 3.

An increase in palpebral aperture also showed a dose–response both at rest and at full smile when BTX-A was used in the lower lid alone. Crow's feet injection demonstrated a synergistic response with the 2 U alone. No such synergy was apparent with 4 or 8 U and crow's feet injection.

Numerical data cannot be used as the sole evidence for cosmetic studies. Our study demonstrates that side effects increased with increasing dose of BTX-A. Patients who were treated with 4 U in the lower lid may need to be advised of some transient dry-eye symptoms because of an increasing exposure of their cornea to the air as a result of the increased palpebral aperture. In addition, clinically, we have noted some cases of lower eyelid edema or lower lid fullness. A small amount of edema may functionally serve to "fill out" lower lid wrinkles and may not be bothersome to the patient. However, an increased amount of lower eyelid edema can produce a worsening of lower eyelid fullness (Figure 5) and give the appearance of increasing lower eyelid "bags." Eight units in the lower eyelid produced significant side effects both cosmetically and functionally, with many patients

reporting that the photophobia, incomplete lid sphincter, and lower eyelid edema were quite bothersome to them. Eight units of BOTOX in the lower lid often produced an undesirable degree of increase in palpebral aperture with patients reporting a "sad" look and exhibiting a good degree of scleral show (Figure 6). There was excess lateral rounding of the lower lid (Figure 7). This is unnatural and unattractive.

The study parallels our clinical experience. Patients who desire improvement of the lower eyelid wrinkles or eye widening are usually initially treated with 2 U placed in the midpupillary line in the lower eyelid along with 12 U in the crow's feet area. The patients are seen back after approximately 2 to 4 weeks, and an assessment is made of their lower eyelid. Should they desire further eyelid widening or improvement in their lower eyelid rhytides, an additional 2 U may be placed into the lower lid. Careful evaluation of the cosmetic effect must be done, looking for lid edema and scleral show. Also, symptoms such as dryness of the eyes or problems with sphincter function of the orbicularis must be assessed before supplementing the patient with an additional 2 U in the lower eyelid. All patients should have an acceptable snap test before treatment. If BTX-A is used solely for eye widening, such as in Asian patients, 2 U can be initially used in each lower lid and a supplemental 2 U added at the 2-week visit. Patients who desire periorbital wrinkle reduction and are not candidates lower eyelid injections due to a poor snap test or scleral show may consider having their lateral eye rhytids treated.

Each patient with periorbital wrinkles needs individual assessment.[17] The best method or combination of methods needs to be appropriately selected for that specific patient. For patients with advanced photodamage laser resurfacing, intense pulsed light treatments, chemical peels, or filler substances may be

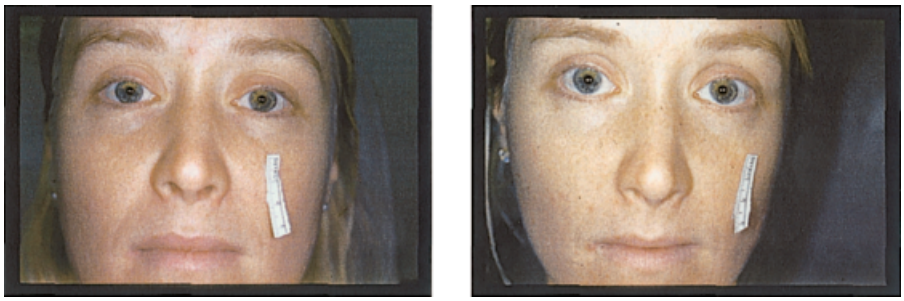


Figure 6. An unattractive appearance is shown with the use of 8 U in the lower eyelid. Scleral show is present producing a "sad" look.



Figure 7. Lower eyelid edema notable after 8 U of BTX-A placed in the lower eyelid. The lateral orbital area was not treated on the patient's right side.

used in addition to BTX-A.[15] A blepharoplasty may be needed in individual patients.

In conclusion, our previous article showed that 2 U in the lower eyelid improved lower eyelid wrinkles and widened the eye. The effect was particularly appreciated by Asian patients. We reported a safe and effective treatment with proper injection technique. Increasing doses of botulinum toxin in the lower eyelid can increase palpebral aperture and decrease lower lid folds. However, these increasing dosages are associated with side effects and undesirable cosmetic changes in the lower eyelid such as scleral show or lower lid edema. Careful assessment must be used in all patients when treating the periocular area. For eye widening, the authors advise use of 2 to 4 U in the lower eyelid. For periocular wrinkles, it is prudent to consider treatment with 2 U in the lower lid along with 12 U in the crow's feet area on initial approach, adding an additional 2 U in the lower lid if needed.

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Commentary

Over the past several years, it has become evident that botulinum A exotoxin is efficacious for temporary reduction of forehead, glabella, and periocular lines. Few studies, however, have focused on side-by-side comparisons of doses, injection sites, and techniques. Given that the toxin is a forgiving material, there is still much to be gained from further optimizing technique. Flynn et al.[1] thus provide a significant service in their lower eyelid dose-finding study.

The use of botulinum toxin under the lower eyelid is perhaps the least well-understood cosmetic application for the upper face.[2] However, this is a site where, when properly used, botulinum toxin can provide a dramatic effect. Not only does suborbital toxin reduce inferior eyelid crows' feet, it also widens and rounds the eye.[3]

Although infraorbital toxin can be highly effective, injecting under the eye requires special care. Sometimes the obtained eye rounding can be excessive, and some patients may complain of looking "funny" or "too different" than their baseline. Sometimes, as the authors note, excessive puffiness can develop under the eye after treatment. Even though the treated muscles are relaxed, this edematous, puffy, bag-like appearance can look worse than the initial lines and creases. Patients with large infraorbital fat pads may be particularly susceptible. Also, for patients with reduced elasticity of the lower eyelid, scleral show, photophobia, and ectropion are possible, and in this subgroup, injections should be avoided. The "snap test" alluded to in the text entails gently tugging down on the lower eyelid and then releasing the eyelid;[4] if the lid only slowly or partially returns to its rest position, this is indicative of decreased elasticity, which should preclude treatment. For all patients, injections under the eye should be placed delicately to avoid bruising. The tiny dermal vessels can be avoided by either multiple vertical punctures or tangential needle entry with injection at several points during withdrawal. If faced with incipient bruising, the injector should immediately apply firm pressure with a gauze and fingertip for several seconds. Injections under the eye should be very superficial because the relevant muscles (orbicularis) are thin and right under the dermis.

These cautions should not dissuade practitioners from injecting toxin under the eye. Usually, this procedure is efficacious and uncomplicated. The authors' recommendations that a total of 2 to 4 U be used for two infraorbital injections are most helpful. Under the eye, as in the crows' feet area,[5] much more toxin does not improve cosmetic effect but does predispose to unwanted effects. For infraorbital injections, the exact best dosage is somewhere between 2 and 4 U per eye: 2 U per eye avoid all unwanted effects with modest cosmetic improvement, but 4 U per eye induce only mild side effects while providing a cosmetic benefit commensurate to that with 8 U per eye. That the main side effect is dry eye suggests that

injections of more than 2 U per eye should be avoided in patients with pre-existing dry eye symptoms or Sjogren's syndrome. Dry eye is most common in middle-aged to older women,[6] and before initial infraorbital injection in such patients, it may be reasonable to elicit relevant history.

Reasonable people can differ over how far below the lash margin injections should be placed. The authors recommend points 3 mm inferior to the ciliary margin, but others inject as much as 1 cm below this landmark. Those injecting relatively high, over the ocular globe, should obviously exercise extra caution regarding the depth of injection.

This extremely helpful dose-finding study[1] will likely establish the benchmark dosage for infraorbital botulinum toxin injections. Also especially useful is the Flynn et al.'s careful itemization of side effects. As we know, botulinum toxin has no long-term side effects, but the transitory inconveniences are best minimized or avoided entirely.

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